Gateway To Your Future

Kuniya College of Management and Technology

Approved by AICTE & Affiliated to
APJ Abdul Kalam Technological University

SCRUM BOOK

20MCA245 – MINI PROJECT(3RD SEM MCA 2025-27)

NAME : Mohammed Irfan KS

Class : S3 MCA

Reg.No: KCMIT25MCA-2011

SCRUM BOOK

Name of the Student:	**Mohammed Irfan KS**
Batch:	**2025-2027**
Register Number:	**KCMIT25MCA-2011**
Academic Year:	**2026**
Subject Name & Code:	**20MCA245 - MINI PROJECT**
Title of the Project	**INTEGRATED ENVIRONMENTAL MONITORING PLATFORM**
Name and Designation of the Guide:	**ATHIRA PJ**

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Vision of the Department

To become a center of excellence in computer applications, competent in the global ecosystem with technical knowledge, innovation with a sense of social commitment.

Mission of the Department

M1:To serve with state of the art education, foster advanced research and cultivate innovation in the field of computer applications.

M2:To prepare learners with knowledge, skills and critical thinking to excel in the technological landscape and contribute positively to society

Program Educational Objectives

PEO I :Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.

PEO II: Graduates will acquire technological advancements through continued education, lifelong learning, and research, thereby making meaningful contributions to the field of computing.

PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively

Program Specfic Outcomes(PSOs)

PSO I: Apply advanced technologies through innovations to enhance the efficiency of design and development

PSO II: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

PO1:Engineering knowledge	PO7:Environment and sustainability
PO2:Problem analysis	PO8:Ethics
PO:3Design/development of solutions	PO9: Individual and team work
PO4:Conduct investigations of complex problems	PO10:Communication
PO5:Modern tool usage	PO11: Project management and finance
PO6:the engineer and society	PO12: Life-long learning

MAPPING

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**2**	**3**	**3**	**3**	**1**	**2**	**3**	**3**	**3**	**3**	**3**	**3**
CO2	**2**	**3**	**2**	**3**	**2**	**3**	**2**	**1**	**3**	**2**	**3**	
CO3	**3**	**3**	**3**	**3**	**3**	**1**	**3**	**3**	**1**		**2**	
CO4	**3**	**3**	**3**	**3**	**3**		**3**	**3**	**1**	**1**	**2**	
CO5	**3**	**3**	**3**	**3**	**3**		**2**	**3**			**1**	
CO6	**3**	**3**	**3**	**3**	**3**	**2**	**3**	**3**		**2**	**3**	**3**
CO7	1	**1**	**3**	**3**	**3**	**2**	**3**	**3**	**2**	**1**	**2**	

MINI PROJECT TOPIC DESCRIPTION

Project Title: Integrated Environmental Monitoring Platform

Project Description:

Integrated Environmental Monitoring Platform is an IoT-based system designed to monitor and analyze indoor microclimate conditions alongside localized outdoor weather data. It uses an ESP32 microcontroller with a DHT22 sensor to capture environmental metrics and transmits them to a cloud-hosted FastAPI backend. The system provides real-time dashboards, dual-trace comparative analytics, ASHRAE 55 thermal comfort scoring, offline persistence, and role-based user access to support climate monitoring.

MODULE DESCRIPTION

1. Microclimate Sensing and Hardware Firmware

This module interfaces the ESP32 microcontroller with a DHT22 sensor and OLED display to collect temperature and humidity readings and transmit telemetry over Wi-Fi.

2. Cloud Ingestion and Weather Sync

This module uses FastAPI to receive sensor data, store records in an SQLite database, and retrieve localized outdoor weather from the Open-Meteo API.

3. Comparative Analytics and Comfort Engine

This module calculates Steadman's Heat Index and evaluates indoor thermal comfort using ASHRAE 55 standards to provide actionable environmental insights.

4. Mobile Dashboard and Visualization

This module provides an interactive Flutter mobile interface with real-time metric cards, synchronized comparative charts, and environmental alert notifications.

5. Local Storage and User Management

This module enables offline data caching using LocalStorageService and provides role-based authentication supporting user profiles and guest access.

WEEKLY REPORT OF THE PROJECT

WEEKLY PROGRESS REPORT

Week: 1

*From:*06/07/2026 *To:* 10/07/2026.

1. Work Done (Should be written by students before meeting your guides)

Project Initiation and Requirement Analysis

The project was initiated by identifying the problem statement and operational requirements of an Integrated Environmental Monitoring Platform. The core system capabilities, including real-time indoor microclimate sensing, cloud-based data ingestion, localized outdoor meteorological synchronization, and cross-platform mobile telemetry visualization, were outlined. The multi-tiered architecture was planned, selecting the ESP32 microcontroller and DHT22 digital sensor for hardware sensing, Python FastAPI for the backend REST API, SQLite for structured storage, and Flutter for the mobile client.

2. Suggestions/Remark by Guide:

Proceed with hardware component selection, circuit schematics, and development environment setup.

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3. Initials by :

GUIDE

WEEKLY PROGRESS REPORT

Week:2

From:13/07/2026 To: 17/07/2026 .

1. Work Done (Should be written by students before meeting your guides)

Hardware Selection and Prototyping Setup

The hardware component specifications were finalized and components were procured, including an ESP32 NodeMCU-32S microcontroller, DHT22 (AM2302) digital temperature and relative humidity sensor, and a 0.96-inch monochrome SSD1306 I2C OLED display. The circuit schematic was designed and prototyped on a breadboard, verifying regulated 3.3V power rails, pull-up resistors for the DHT22 single-bus data line, and I2C SDA/SCL pin assignments (GPIO 21 and 22). Baseline firmware tools were configured in the Arduino IDE with necessary board definition packages.

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2. Suggestions/Remark by Guide:

Verify sensor accuracy, I2C address resolution, and establish stable pinout connections.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week:3

From:20/07/2026 To: 24/07/2026 .

1. Work Done (Should be written by students before meeting your guides)

Sensor Interfacing and Local Display Implementation

Firmware was developed in C++ to interface the ESP32 with the DHT22 sensor and SSD1306 OLED display using Adafruit DHT and SSD1306 libraries. An I2C scanner routine was executed to confirm display communication at address 0x3C. The firmware logic was implemented to sample temperature and humidity values at non-blocking 2-second intervals, format the readings, and render real-time indoor environmental metrics on the OLED screen. Preliminary bench testing verified reading consistency and display responsiveness.

2. Suggestions/Remark by Guide:

Implement non-blocking sampling intervals and prepare Wi-Fi communication routines.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 4

From:27/07/2026 To: 31/07/2026

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1. Work Done (Should be written by students before meeting your guides)

ESP32 Wi-Fi Integration and Telemetry Serialization

The edge firmware was expanded to include Wi-Fi networking using the ESP32 WiFi library. An automated connection handler was implemented to authenticate against local access points and manage automated reconnection routines upon network interruption. The ArduinoJson library was integrated to structure temperature, relative humidity, computed heat index, and device status into serialized JSON telemetry payloads. Local network HTTP client routines were established to prepare for cloud transmission.

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2. Suggestions/Remark by Guide:

Design the backend REST API specifications and database models for telemetry ingestion.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 5

From.:03/08/2026 To: 07/08/2026

1. Work Done (Should be written by students before meeting your guides)

Backend Architecture and Database Design

The cloud backend structure was developed using Python FastAPI and the Uvicorn ASGI framework. The relational database schema was modeled using SQLAlchemy ORM to manage telemetry tables, capturing sensor timestamps, temperature, relative humidity, computed heat index, and hardware status flags. An SQLite database was initialized with structured time-series indexing. Pydantic data schemas were defined to enforce strict request validation, payload typing, and robust error handling for incoming IoT telemetry endpoints.

2. Suggestions/Remark by Guide:

Proceed with API endpoint development and test telemetry ingestion with simulated requests.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 6

From.:10/08/2026 To: 14/08/2026

1. Work Done (Should be written by students before meeting your guides)

Ingestion API Development and Testing

RESTful API endpoints for telemetry ingestion (/api/telemetry) and retrieval (/api/telemetry/latest, /api/telemetry/history) were implemented in FastAPI. The endpoints were tested using Swagger UI and Postman to ensure reliable JSON parsing, automated UTC timestamp generation, and efficient database insertion. Data sanitization routines were integrated to validate sensor value ranges and filter abnormal readings. Query response times for time-series records were tested and optimized.

2. Suggestions/Remark by Guide:

Integrate the external meteorological weather service to provide outdoor reference data.

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3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 7

From.:17/08/2026 To: 21/08/2026

1. Work Done (Should be written by students before meeting your guides)

External Meteorological API Integration

An external meteorological synchronization pipeline was developed in the FastAPI backend utilizing the Open-Meteo REST API. A background task was created to query localized outdoor meteorological data based on geographic coordinates, capturing ambient temperature, relative humidity, wind speed, and precipitation probability. The backend logic was arranged to align outdoor reference data alongside indoor sensor telemetry, enabling synchronized dual-source environmental analysis.

2. Suggestions/Remark by Guide:

Develop the comparative analytics algorithms and thermal comfort evaluation models.

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WEEKLY PROGRESS REPORT

Week: 8

From.:24/08/2026 To: 28/08/2026

1. Work Done (Should be written by students before meeting your guides)

Thermal Comfort and Comparative Analytics Engine

Analytical computation modules were developed to derive meaningful environmental insights from indoor and outdoor datasets. The Steadman Heat Index formula was implemented to compute apparent ("feels like") temperature accounting for humidity-driven evaporative cooling reduction. Thermal comfort boundaries were modeled based on ASHRAE Standard 55 envelopes, classifying indoor conditions into "Optimal", "Acceptable", or "Discomfort" states. Indoor-outdoor differentials were calculated to evaluate building thermal retention and HVAC efficiency.

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2. Suggestions/Remark by Guide:

Initialize the mobile application project and design the primary user interface layout.

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WEEKLY PROGRESS REPORT

Week: 9

From.:31/08/2026 To: 04/09/2026

1. Work Done (Should be written by students before meeting your guides)

Mobile Application Setup and Architecture

The cross-platform mobile client application was created using Google Flutter 3 and Dart SDK. The application architecture was established using the Provider pattern to enable scalable state management and reactive UI updates. Core project folders for screens, models, services, providers, and widgets were structured. Theme data featuring clean modern typography and responsive card layouts was configured, and basic screen routing was set up and tested on an Android emulator.

2. Suggestions/Remark by Guide:

Implement HTTP network service and bind live telemetry data to dashboard cards.

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3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 10

From.:07/09/2026 To: 11/09/2026

1. Work Done (Should be written by students before meeting your guides)

Mobile Dashboard and Telemetry Cards

The primary dashboard interface was developed in Flutter to present live environmental telemetry. Responsive metric cards were implemented to display real-time indoor temperature, relative humidity, heat index, and thermal comfort category status. An HTTP client service was integrated to fetch telemetry updates from the backend REST endpoints. Visual indicators, color-coded badges, and icons were arranged to visually represent environmental quality and status levels.

2. Suggestions/Remark by Guide:

Develop interactive graphical charts to visualize comparative telemetry trends over time.

To implement better security and option to get an overall rating in the show details page

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 11

From.:14/09/2026 To: 18/09/2026

1. Work Done (Should be written by students before meeting your guides)

Comparative Chart Visualization Development

Interactive time-series graphical charts were integrated into the Flutter application utilizing the fl_chart library. Dual-trace line graphs were implemented to render indoor sensor telemetry directly alongside localized outdoor Open-Meteo meteorological curves. Interactive tooltips, touch markers, and customized grid lines were configured. Timestamp parsing routines were developed to convert UTC cloud records into local IST (UTC+5:30) time to ensure accurate chart timeline alignment.

2. Suggestions/Remark by Guide:

Implement on-device local storage persistence to enable offline access and historical auditing.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 12

From.:21/09/2026 To: 25/09/2026

1. Work Done (Should be written by students before meeting your guides)

Local Storage and Offline Data Persistence

A local persistence layer was developed using LocalStorageService and SharedPreferences in Flutter. The service was engineered to cache recent telemetry records in persistent JSON format on device storage upon receipt. Logic was implemented to check network connectivity, allowing the mobile application to seamlessly load cached historical records, view past trends, and maintain dashboard functionality when disconnected from Wi-Fi or backend services.

2. Suggestions/Remark by Guide:

Implement user authentication, role-based access control, and climate alert notifications.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 13

From.:28/09/2026 To: 1/10/2026

1. Work Done (Should be written by students before meeting your guides)

User Management and Alert Engine

Role-Based Access Control (RBAC) was implemented in the Flutter application, supporting authenticated user profiles ("User: Irfan") and an unrestricted guest mode for open telemetry monitoring. A session management service was integrated to preserve login state across app restarts. Additionally, an intelligent environmental alert system was created to monitor humidity thresholds and thermal comfort boundaries, displaying single relevant alerts for current climate conditions while preventing alert duplication.

2. Suggestions/Remark by Guide:

Deploy the backend to cloud infrastructure and conduct end-to-end integration testing.

3. Initials by :

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WEEKLY PROGRESS REPORT

Week: 14

From.:05/10/2026 To:09/10/2026

1. Work Done (Should be written by students before meeting your guides)

Cloud Deployment and End-to-End Edge-Cloud Testing

The FastAPI backend service was deployed to Render cloud hosting (https://environmental-monitoring-backend.onrender.com) with persistent database storage. The physical ESP32 edge node was programmed with production cloud URLs and tested under live conditions, reliably transmitting DHT22 sensor readings over Wi-Fi. End-to-end data transmission, cloud ingestion, Open-Meteo weather synchronization, and mobile dashboard updates were validated across sustained continuous operating cycles.

2. Suggestions/Remark by Guide:

Perform physical device verification, final optimizations, and prepare presentation artifacts.

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WEEKLY PROGRESS REPORT

Week: 15

From.:12/10/2026 To:16/10/2026

1. Work Done (Should be written by students before meeting your guides)

Physical Device Testing, Refinement and Final Verification

The mobile application was compiled into a release APK and installed on a physical smartphone (Samsung Galaxy A55 5G) for real-world field testing. UI responsiveness, app icon rendering, real-time graph synchronization, and offline cache retrieval were verified under actual operating conditions. Minor display and font styling adjustments were refined. The entire system architecture was finalized, and the mini-project documentation and Scrum Book were compiled for evaluation and viva voce.

2. Suggestions/Remark by Guide:

The project is complete, fully validated, and ready for demonstration. Documentation approved for final submission.

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